

Quasi-Static Laptime Simulation

May 2014

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Overall Simulator

The quasi-static simulator discretizes track segments and assumes that the car is an equilibrium state in each segment. The car can then be in a combined state of lateral and longitudinal accelerations using D'Alembert's principle . The simulator does not take into account any suspension kinematics since it is analyzing a kart and the tire model is too simple.

First, the user must open the Matlab file Laptime.m. Then the user must enter a set of car parameters. The next table shows these parameters.

Table 1-Kart Parameter

Car parameter	Matlab variable name	Value
Mass	m	220 kg
Mass repartition (front)	fm	0.46
Front track width	t_f	1.1 m
Rear track width	t_r	1.2 m
Wheel base	l	1.18m
CofG height	h	0.125 m
Tire radius	tire_r	0.143 m
Frontal area	f_area	0.91 m ²
Coefficient of drag	cd	0.46
Aero efficiency	aero_eff	1.5
Front roll moment distribution	roll_dist	0.5
Front Aero Load distribution	cp_f	0.4
Brake bias front	brake_bias	0.6
Optimal slip angle	slip_angle	3 deg

The power train parameters are also entered. Here is the gearbox, engine map and tire data

RPM	Power [Hp]
8000	25
9000	40
10000	60
11000	70
12000	85
13000	90
14000	80
14000	Rev-limiter

Gear	Ratio
1	2.5
2	2
3	1.571
4	1.353
5	1.222
6	1.15
Final_Drive	2.9

Normal force [N]	Available grip [N]
250	300
500	800
750	1200
1000	1400
1250	1350
1500	1250

Then some simple simulation parameters are entered to make the simulation faster but less accurate or the opposite.

Table 2-Simulation Parameters

Simulation parameter	Value	Description
curve_threshold	0.002 m ⁻¹	Minimum value of curvature that is considered a curve not a straight line
frac	0.1	The fraction of the Nyquist frequency used to create the Butterworth filter
n	2	Order of the Butterworth filter
min_peak	0.005 m ⁻¹	Minimum curvature to be considered as an apex
d_step	2 m	Distance step discretizing the track
v_gear_change		Vehicle speed at which the gear change must be for every gear
v_step	0.01 m/s	Speed steps in combined states
q	5	Number of Cd value tested
min_cp	0.2	Minimum Cd value tested
max_cd	0.5	Maximum Cd value tested
ii	6	Number of pressure distribution tested
min_cp	0.3	Minimum pressure distribution
max_cp	0.5	Maximum pressure distribution

Track Map

The kart's velocity, lateral acceleration and distance travelled channels were imported to Matlab to create the track map using a TXT file. To create the map, the curvature profile of the track must be calculated. Using the other channels listed above, the curvature is calculated as follow:

$$Curvature = \frac{lateral\ acceleration}{speed^2}$$

Using the raw data would result in a very noisy track map. Therefore a Butterworth filter on the curvature was implemented. Also the curvature is interpolated at a constant distance step to simplify the simulation. The order of the filter and the fraction of the Nyquist frequency are used to fine-tune the filter. The aim of the filter is to remove high frequency noise without altering the amplitude and the phase of the low-frequency data. It is important that double-apex corners still appear on the track map. Having a too low cut-off frequency would lead to a track map that does not represent at all the real track. Figure 1 shows this phenomenon with a cut-off frequency of 1% of the Nyquist frequency.

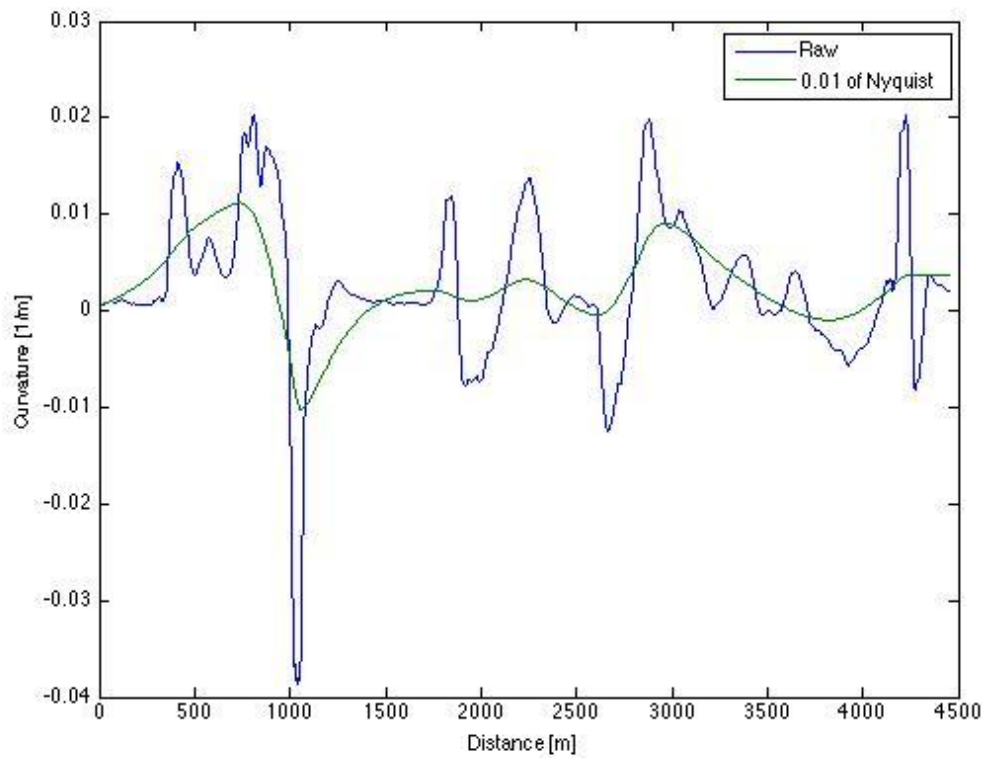


Figure 1-Effects of low cut-off frequency on curvature

On the other hand, having a high cut-off frequency will not filter enough the data and the curvature trace will not be smooth enough.

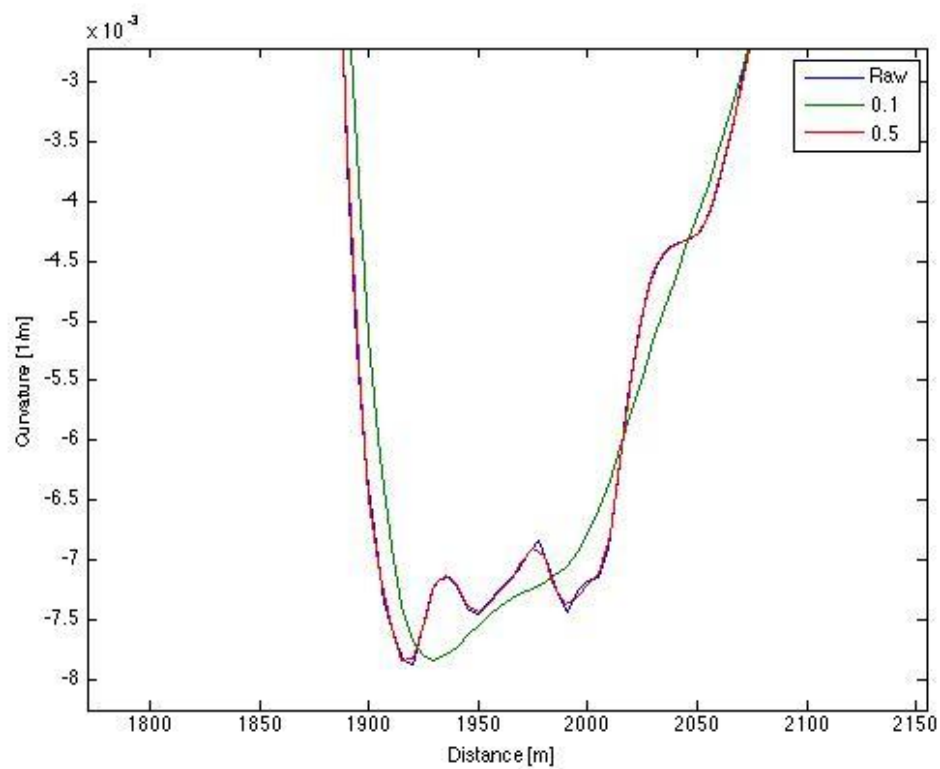


Figure 2- Effects of cut-off frequency on curvature

Figure 2 displays the data filtered with 50% and 10% of the cut-frequency on a specific corner. The 50% curve follows almost perfectly the raw data. The 10% curve however, follows the trend but does not exhibit the high frequency noise of the raw data. The magnitude is slightly affected and the phase lag is minimal. Therefore a 10% cut-off frequency was chosen.

There are no clear trends using different orders of filter other than computing power necessary. It was found that the second order filter was well suited for the task.

Figure 3 show the track map generated with the filter. An algorithm is also used to find the apexes. The algorithm finds local minimum and maximum, which have an absolute curvature than the min_peak value (0.005 m^{-1}). The value was determined by making sure the track map had eighteen apexes like the real track shown on Figure 4.

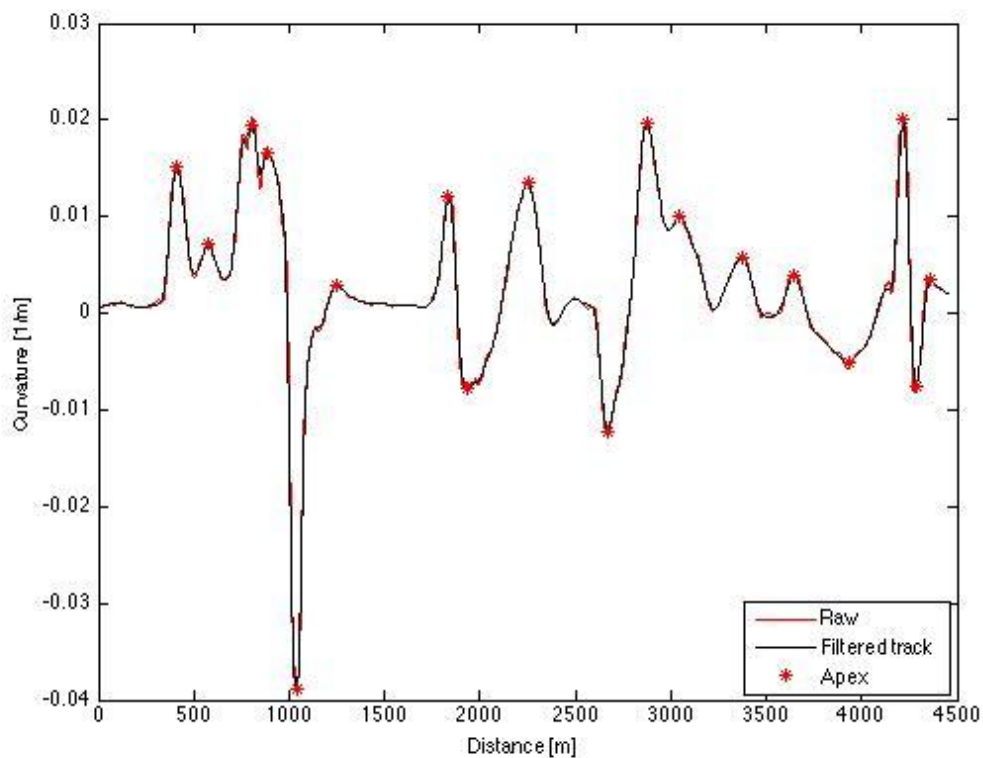


Figure 3-Generated track map Vs raw track map



Figure 4-Assen track layout

Kart model

Once the track has been defined, the simulation calculates the maximum speed achievable at every apex. This is done by assuming that the kart is in a pure cornering state ie no longitudinal acceleration. Then the simulation starts from the slowest corner and calculates the maximum speed achievable at constant distance steps in a combined state between two apexes. This will be explained later.

Gearbox

Using the data on track and the setup sheet, the gearbox ratios were found. Using the fifth gear, it was possible to find the final gear ratio and then all the other ratios. Figure 5 shows an overlay of the raw data of the kart and the gearbox ratio used in the simulation. On the setup sheet, the first gear ratios available did not match the one calculated using the raw data. The simulation uses the one calculated from the track data. Here are the equations used to determine the ratios:

$$\text{Wheel rpm} = \text{velocity} * \frac{60}{2 * \pi * \text{tire radius}}$$

$$\text{Final drive} = \frac{\text{engine rpm}}{5\text{th gear ratio} * \text{wheel rpm}}$$

$$\text{Gear ratio} = \frac{\text{engine rpm}}{\text{wheel rpm} * \text{final drive}}$$

The ratios never exactly match the ones available on the setup sheet but were approximated with the closest ratio available.

Table 3-Gearbox ratios

1	2	3	4	5	6	Final Drive
2.5	2	1.571	1.353	1.222	1.15	2.9

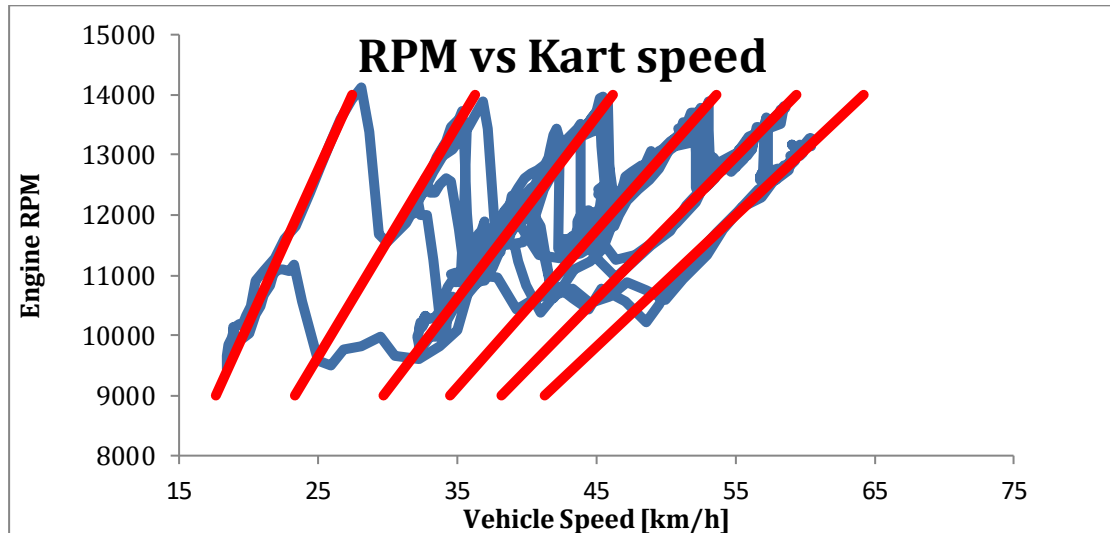


Figure 5-Gearbox ratios identification

Pure cornering for maximum achievable apex speed

The pure cornering phase compares the speed at which each axle saturates and takes the smallest one. The model takes into account aerodynamic loads, static loads, and lateral weight transfer.

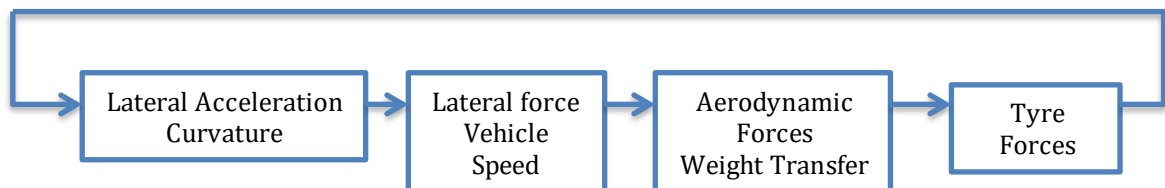


Figure 6-Flow Chart of the Pure Cornering Algorithm

The equations used in the model are listed below

- $Lateral\ Force = m * Lateral\ acceleration * \%of\ mass\ on\ the\ axle$
- $Drag = 0.5 * cd * density * v^2 * frontal_area$
- $Total\ down\ force = Drag * aero_{efficiency}$
- $Wheel\ loads = static\ load + m * h * lateral\ accel * \frac{track\ width}{l} * roll\ distribution + aero\ load\ on\ axle / 2$
- With the all the wheel loads, the lateral force produced by each axle is calculated with the tire data and a linear interpolation.
- $Velocity = \sqrt{(Lateral\ Force) / (m * \%of\ mass\ on\ the\ axle * curvature)}$

The algorithm is done on each axle and stops when the new calculated lateral force is smaller than the previous one.

The chassis model does not take into account kinematics, transient weight transfer and supposes an infinitely stiff chassis with a given roll moment distribution that stays constant.

Powertrain

Using the engine data, a torque curve with respect to vehicle speed is done using a cubic interpolation as shown in Figure 7.

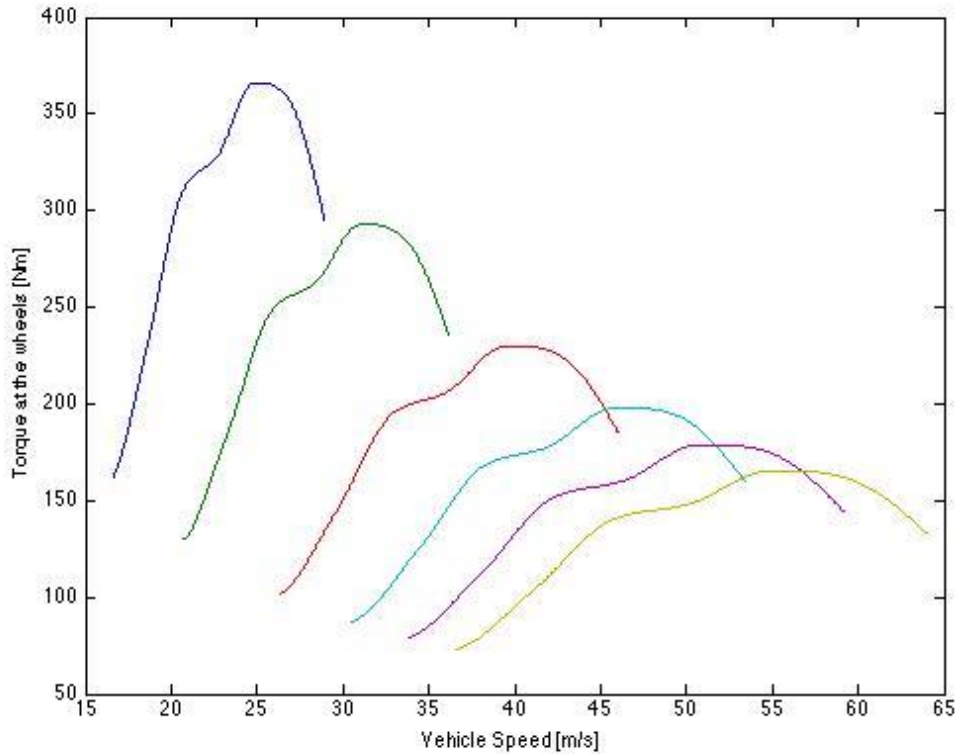


Figure 7-Torque at the Wheels vs Vehicle Speed

With this curve, the user can then define the shifting speeds, which are the intersecting point of two curves or the speed corresponding to the rev-limiter. The engine force at the wheel profile is then calculated and shown in the next graph.

If the engine rpm is smaller than the lowest rpm of the engine map, the model does a linear interpolation of the engine torque between 0 and the torque available at the lowest given rpm. The following equations were used in the model.

- $Engine\ rpm = wheel\ rpm * gear\ ratio * final\ drive$
- $Engine\ Torque = engine\ power\ (at\ the\ specific\ rpm) * \frac{60}{engine\ rpm * 2 * \pi}$
- $Force\ at\ the\ wheel = \frac{engine\ torque * gear\ ratio * final\ drive}{tyre\ radius}$
- $Vehicle\ Speed = engine\ rpm * 2\pi * tyre\ radius / (gear\ ratio * final\ drive)$

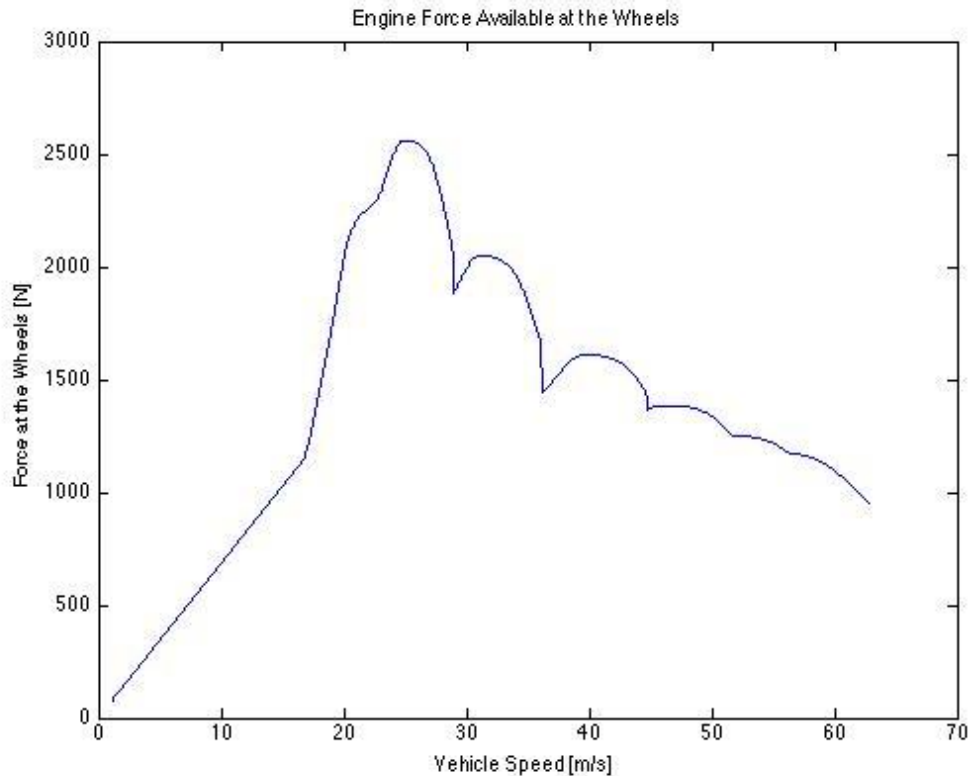


Figure 8- Engine Force Available at the Wheels vs Speed

Aero and Tyres

The aerodynamic forces are divided as drag and downforce. They are calculated with the well know formulas.

- $Drag = 0.5 * cd * density * v^2 * frontal_area$
- $Total\ down\ force = Drag * aero_{efficiency}$

As for the downforce, it is distributed just like mass is with the center of gravity but this time with center of pressure. So if the center of pressure is 40% front, then the downforce on the front axle is 40% of the total downforce.

As for the tyre, the curve provided represents the total force as a function of vertical load. This means that the force calculated represents the radius of the traction circle. The tyre curve is shown in the next figure.

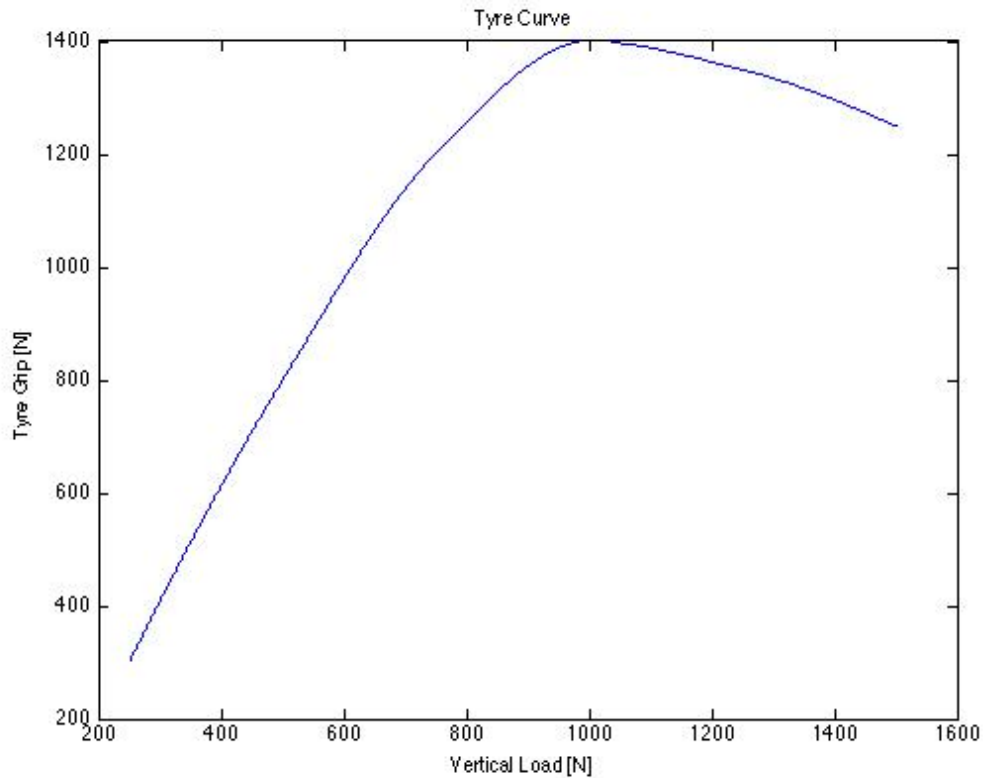


Figure 9-Tyre Curve

The combined model

The combined model assumes that the kart is in an equilibrium using D'Alembert's principle. The kart is under the effect of lateral and longitudinal accelerations.

The track has been discretized into constant distance steps with an associated curvature. The apexes have been found and the maximum achievable apex speeds have been calculated using the pure cornering model. Constant longitudinal and lateral acceleration are assumed over the distance steps.

The simulation must start from the slowest corner. It calculates the maximum achievable speed from the point that is at a distance step away from the apex all the way to the next apex. Then it starts from the second apex and calculates backwards the maximum speed possible that the point before could have in order to be able to brake down to that speed. This is done for every step between the two apexes. Then, the intersection point is found to define the braking point. This is shown Figure 10 where the red line is the acceleration phase between the first and the second apex and the blue line, which is calculated from the second apex to the first apex, shows the braking phase. Then the two curves are combined to give the velocity profile between both apexes as seen on Figure 11.

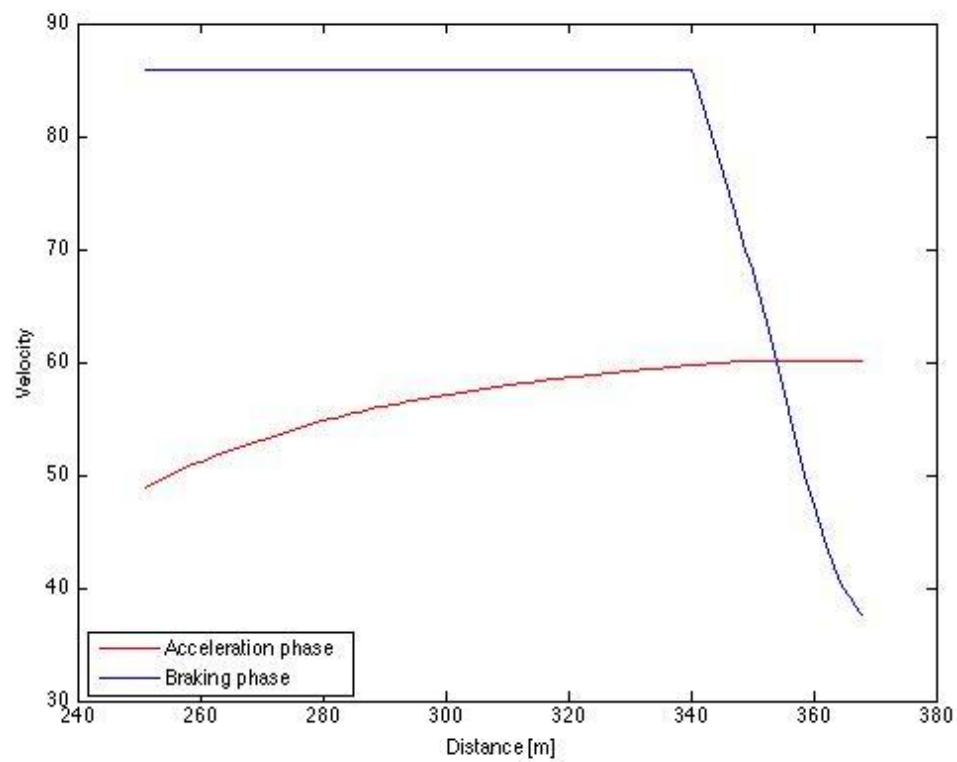


Figure 10-Acceleration and Braking Phase

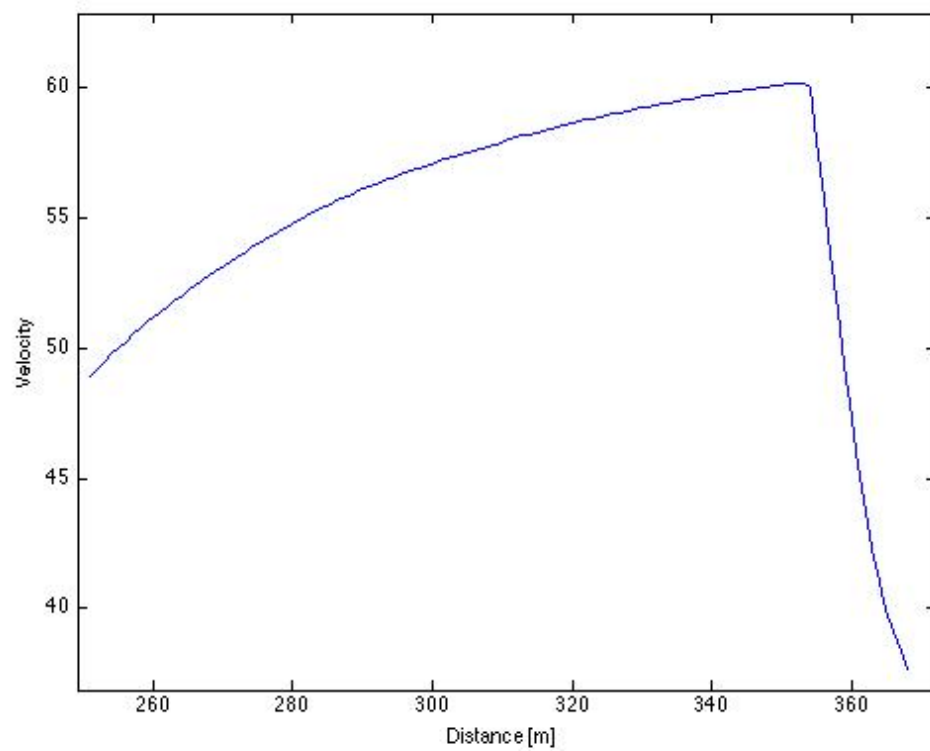


Figure 11- Velocity Profile Between Two Apexes

The acceleration algorithm is basically identical to the braking one. The only two exceptions are that in the acceleration phase, only the rear tyres are affected by longitudinal forces, while in the braking phase, all four tyres have longitudinal forces acting on them and that in the braking phase, the aerodynamic drag helps the vehicle to decelerate, while in the acceleration phase, it counter-acts the engine forward force. These are the reasons why the kart can brake much harder than it can accelerate. Figure 12 shows the thinking behind the combined model.

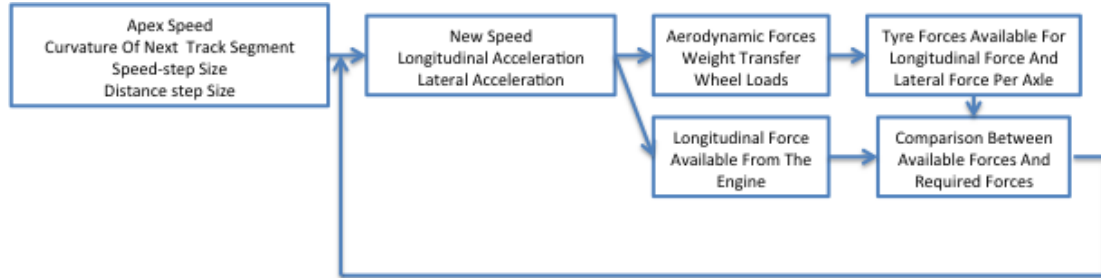


Figure 12-Combined Model Flow Chart

The model uses speed steps. Knowing the track segment length and the initial speed (given by the speed of the previous segment) and the final speed, the longitudinal acceleration is calculated. The final speed is also used to calculate the lateral acceleration needed to go through that track segment at this final speed with the given curvature. The aerodynamic forces, the weight transfer and the wheel loads are then calculated. Then the available forces are compared to the required forces on each axle to determine if the car is power-limited, traction limited or if there is still tyre grip available. If tyre grip is still available, then the speed is increased by one speed increment. The following equations are used for the combined model.

- $Longitudinal\ acceleration = (v_{final}^2 - v_{initial}^2) / (2 * distance_step)$
- $Lateral\ acceleration = v_{initial}^2 * curvature$
- $Longitudinal\ weight\ transfer = m * h * longitudinal\ \frac{acceleration}{l}$
- $Required\ longitudinal\ force = mass * longitudinal\ acceleration + drag$
- $Required\ lateral\ force\ on\ each\ axle = mass * \%\ of\ mass\ on\ axle * lateral\ acceleration$

Using the force at the wheel curve (Figure 8) the available force at the wheel from the engine is defined.

The power limited condition is defined as:

$$If\ force\ available\ from\ the\ engine < (required\ longitudinal\ acceleration + tyre\ drag)$$

then the kart is power limited. The tyre drag is defined as:

$$Tyre\ drag = lateral\ force\ required * \sin(slip\ angle)$$

where the slip angle is defined by the user. It is used to reduce the top speed in high speed corners.

The grip limited condition is defined as :

$$\begin{aligned} & \text{tyre grip available at the front axle} \\ & \leq \text{mass} * \% \text{mass on front axle} * \text{lateral acceleration} \\ & \text{AND} \\ & \text{tyre grip available at the rear axle} \\ & \leq \sqrt{(\text{mass} * \% \text{mass on rear axle})^2 + (\text{required longitudinal force at the axle})^2} \end{aligned}$$

For the deceleration phase, the brake-bias parameter was introduced to divide between the front and the rear axle the longitudinal force required to brake the car. The following equation shows the condition to verify if the car is traction limited while braking:

$$\begin{aligned} & \text{tyre grip available at the front axle} \\ & \leq \sqrt{(\text{mass} * \% \text{mass on front axle})^2 + (\text{required longitudinal force} * \text{brake bias})^2} \\ & \text{AND} \\ & \text{tyre grip available at the rear axle} \\ & \leq \sqrt{(\text{mass} * \% \text{mass on rear axle})^2 + (\text{required longitudinal force} * (1 - \text{brake bias}))^2} \end{aligned}$$

These conditions are based on axle traction circle. This means that the grip available from each tyre is calculated and summed for both axles. This implies that the model does not take into account the yaw moment created by having only one wheel accelerating the car forward.

Validation

The simulation is validated against the fastest lap but it simulates a perfect lap. Therefore, the lap times calculated by the simulation should be smaller than the actual one. By combining all the best reasonable sector times of the six laps available, the perfect lap is estimated to be around 96 seconds but could also be smaller if a better driver was available.

The size of the speed increments was set to 0.001 m in order to give a good resolution in the acceleration phase. Then, the distance step size was studied to see how it effects the simulation. Distance steps of 5 meters were used as it reduces the computing time without affecting the lap time too much. The important point here is that we want lap times in the right range and we are less concerned by the absolute value but more by the trends and the differences between two kart setups. The minimum speed on track is about 20 m/s and recorded at 10Hz, which gives 2 meters as the smallest distance step between two data points. Therefore using 2 meter increments seems to be a good choice.

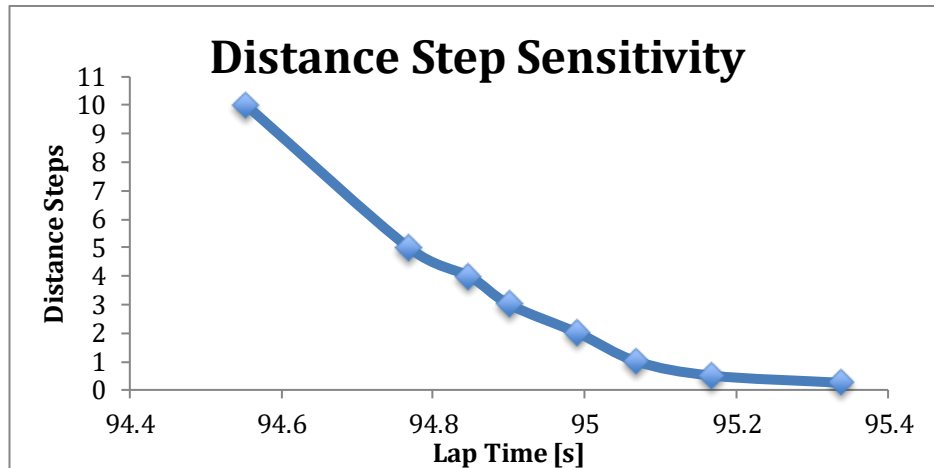


Figure 13-Distance Step Sensitivity Analysis

Local grip factors were used to match apex speeds with the real data. This was done by adjusting the curvature at the apexes. However, this added some discontinuities in the curvature profile and results were poor. Therefore, the grip factors were applied to every curvature point from half of the distance between the previous apex and the apex all the way to half of the distance between the apex and the next one. Considering that between two apexes, there is a straight line (not true for all corner but for most), then varying the curvature of a straight line does not greatly affect the performance of the kart on it. The user can also change which apex he wants that feature to be applied. For this report, all apexes use that characteristic.

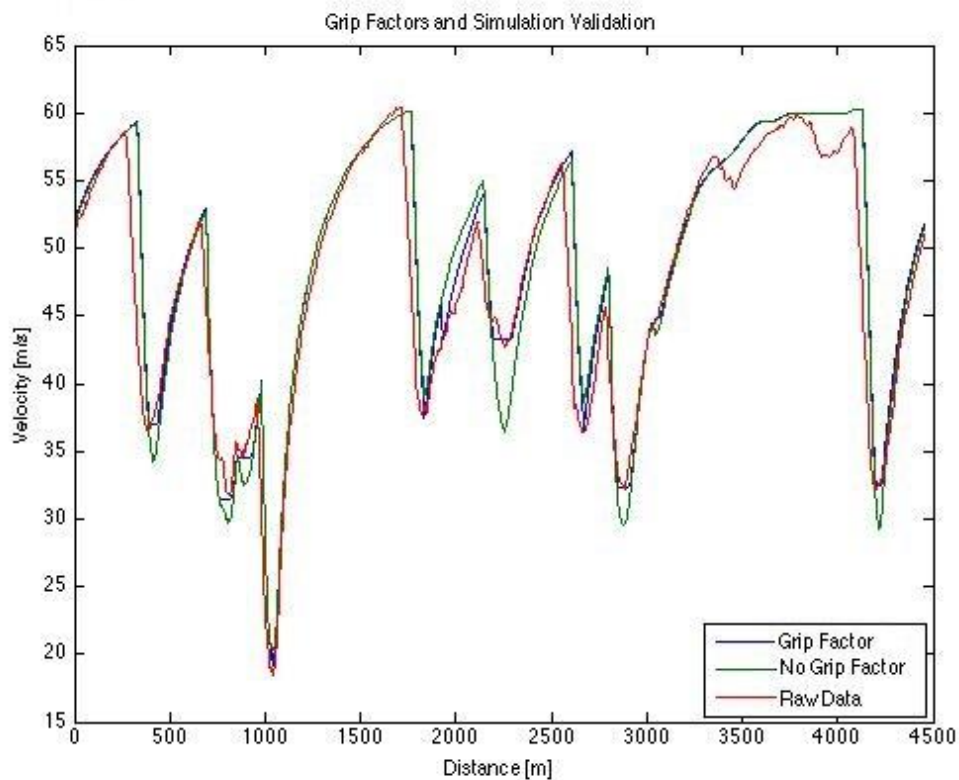


Figure 14- Grip Factor and Simulation Validation

Figure 14 shows a good correlation between the raw data and the simulation. Most of the discrepancies are in the last section of the track where the driver seems to scared to go fast. Figure 15 shows the simulation against multiple laps.

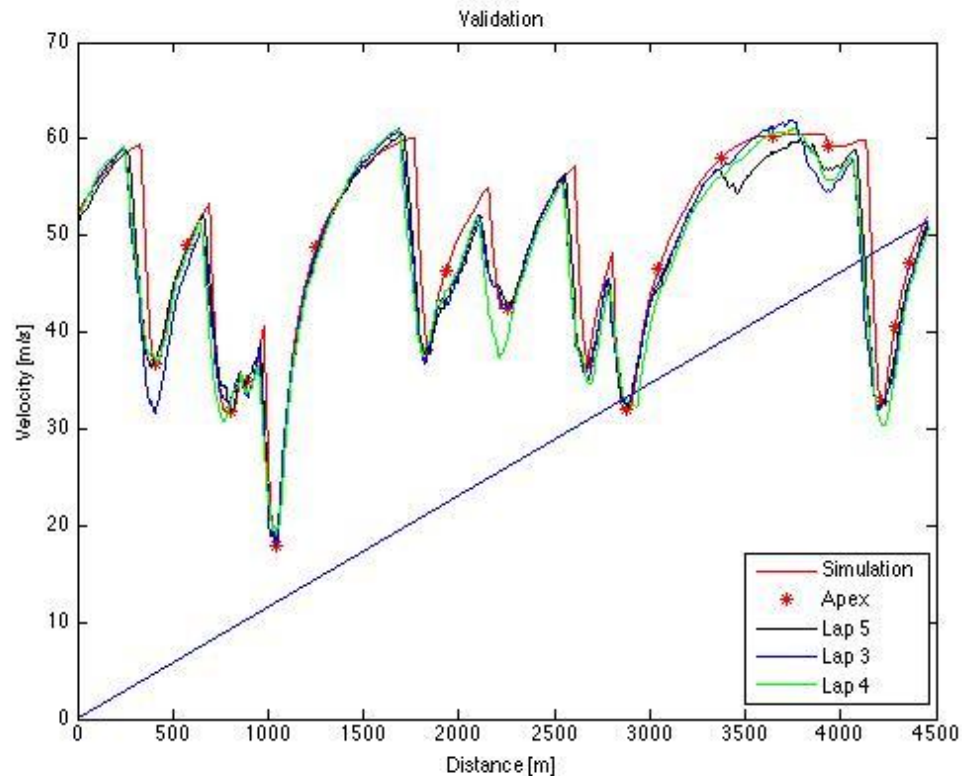


Figure 15-Validation Against Multiple Laps

The major discrepancies are found at apex 8 where the driver seems not to accelerate as fast as he could and in the last portion. Since the simulation is trying to simulate a perfect lap and because the most of the simulation shows very good correlation, these discrepancies are accepted. Let's not forget that the absolute value is not as important as the trends. The aerodynamic simulation study is now ready.

Results

The task was to identify the best aerodynamic configuration possible with the kart. The aerodynamic efficiency was kept constant as it is obvious that increasing downforce while reducing drag would always be beneficial. The effects of coefficient of drag and the center of pressure location were studied. As the simulation is time consuming, only 651 combinations were tested. The study could be refined easily by changing two parameters to test more combinations.

After very rough tests, it was determined that the ideal drag coefficient is in the low 0.30 and that the center of pressure location was somewhere around the 0.4. Therefore, the drag coefficient was varied from 0.24 to 0.37 and the center of pressure location was varied from 0.35 to 0.5. Figure 16 shows the result of the study. The optimal combination found is a drag coefficient value of 0.292 with a

center of pressure location of 0.435. This reduces the lap time to 93.86 seconds compared to 94.94, which is a massive gain.

Figure 17 shows how the optimal center of pressure location varies with drag coefficient. As the drag coefficient increases, the optimal center of pressure location tends to be more forward.

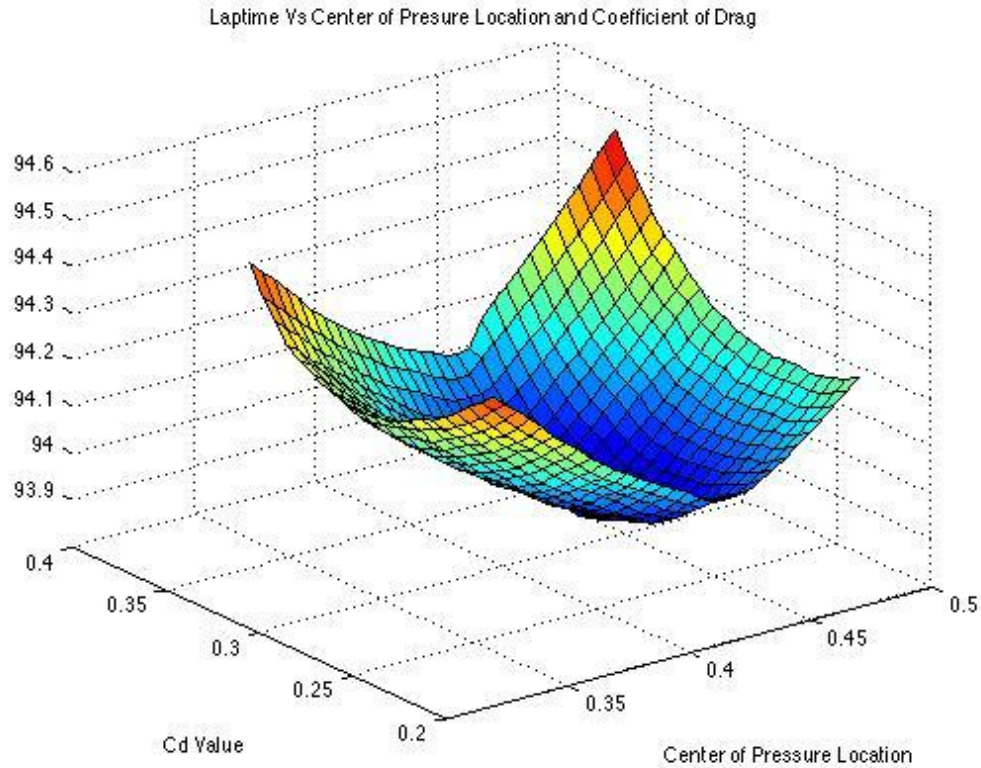


Figure 16-Aerodynamic Parameter Study

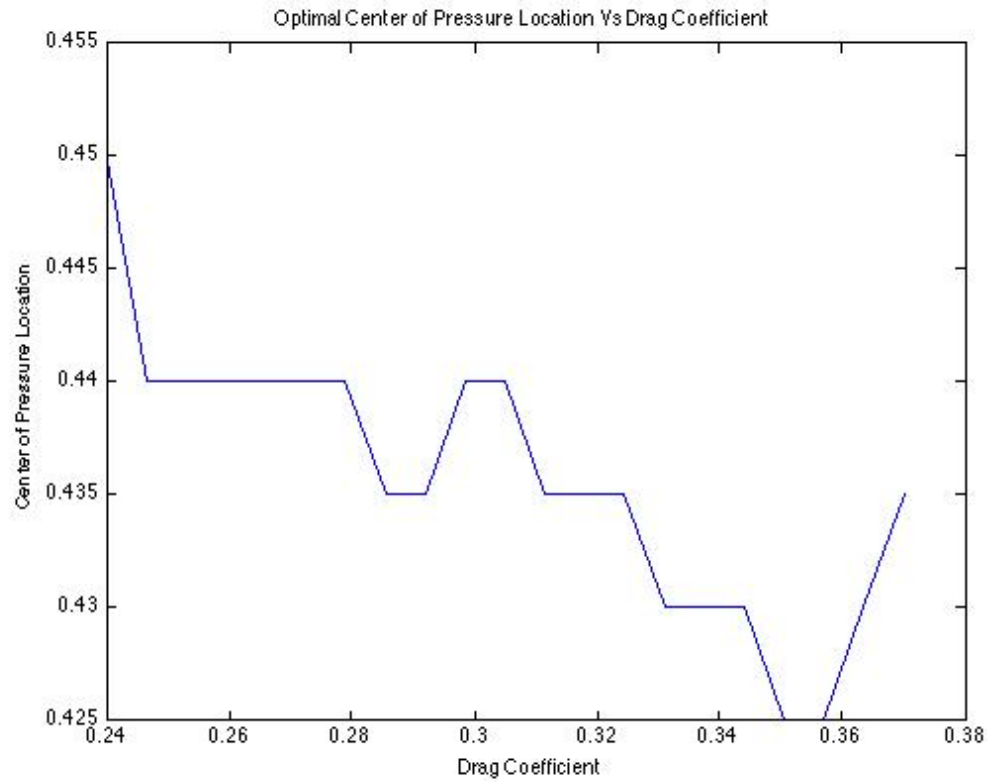


Figure 17-Optimal Center of Pressure Location Vs Drag Coefficient

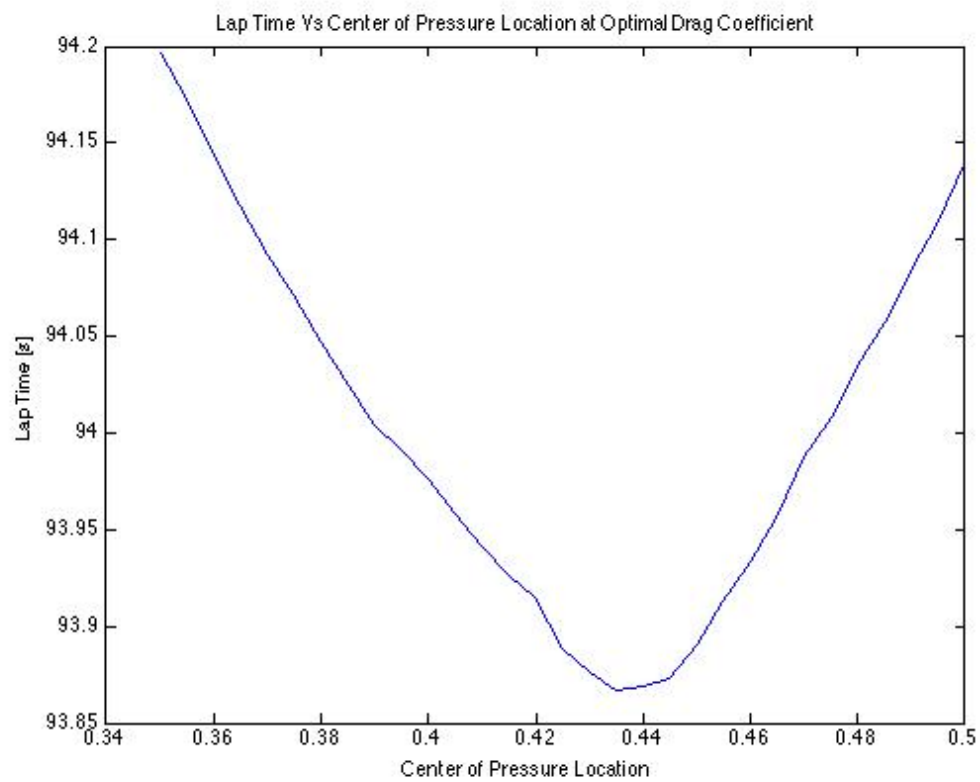


Figure 18-Lap Time Vs Center of Pressure Location at Optimal Drag Coefficient

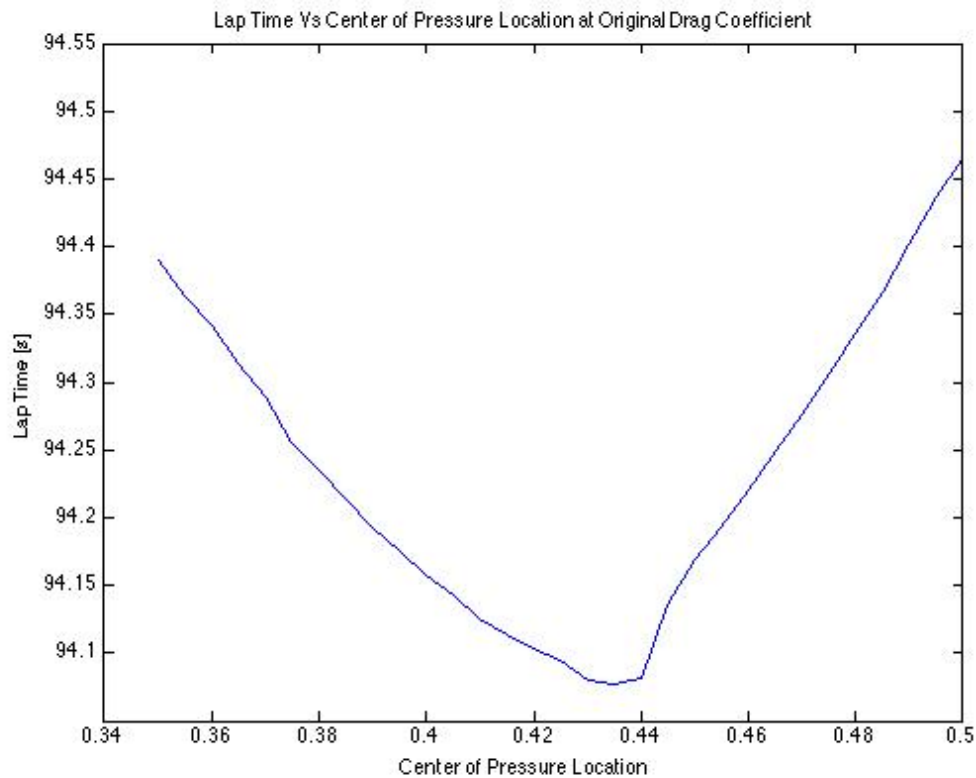


Figure 19-Lap Time Vs Center of Pressure Location at Original Drag Coefficient

From Figure 18 and Figure 19, one can see how the center of pressure location affects lap times with the optimal drag coefficient and the original respectively. It shows that the team should bring the center of pressure a bit more towards the rear in both cases.

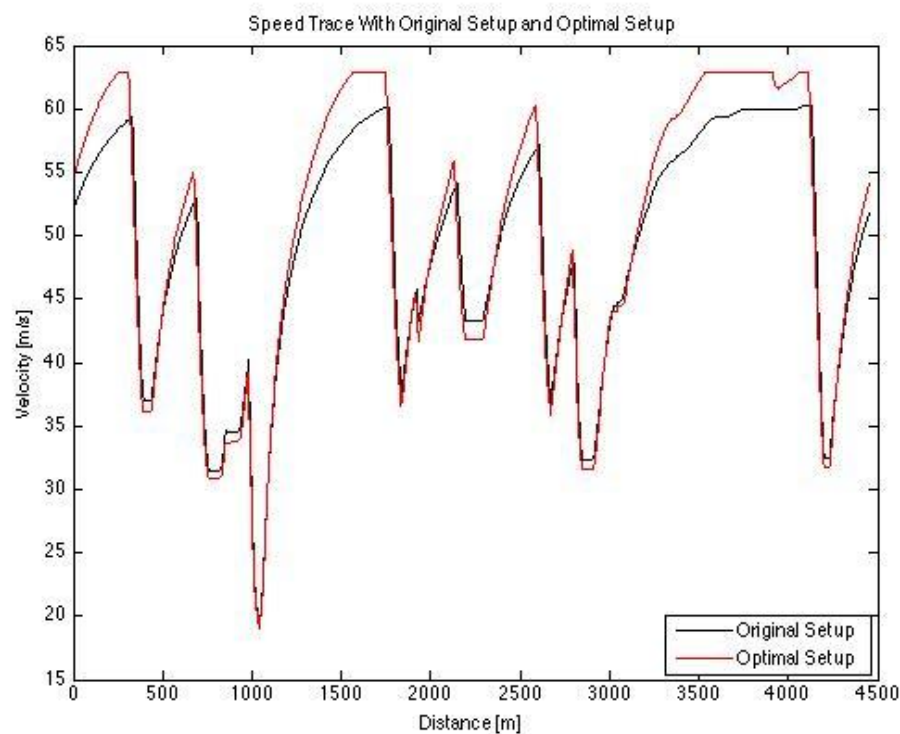


Figure 20-Speed Trace Comparison with Optimal and Original Setup

On Figure 20, the differences between both setups is easy to see. As expected, with the smaller amount of downforce, the cornering speeds of the new setup are smaller and top speed is much greater. On a high-speed track, it is often worth to decrease the drag coefficient to gain speed in the straights at the expense of lower cornering speeds to reduce lap times.

We can also see that the gearbox ratios should be changed with the new setup to make the last gear longer as a plateau is reached in the long straights.

This simulation is very simple but enables the user to perform sensibility analysis on multiple parameters such as:

- engine power
- tyres
- drag
- aerodynamic efficiency
- aerodynamic center of pressure location
- center of gravity location
- mass
- gearing
- track width
- roll couple distribution

and help the engineer to better understand the contribution of every parameter without having the real absolute value of the change made.

Future work

Working on a simulation is a never ending process. Vehicle model and track model refinements can always be added.

The track model could have a grip factor at every distance step with a smoother variations. It could also include banking angle, bumps, grade and even wind speed.

The car model could have an aero map, kinematics, compliances, camber gain curves, bump-steer curves and the slip ratio and slip-angles would need to be calculated. The tire model would need to take into account combined slip conditions and inclination angle. This way it would be possible to do some tuning of the car before the going to the venue or at least have an idea of the sensibility of each suspension parameters.

A fully transient model could also be developed to take into account transient weight transfer, transient tire model and also damping forces acting on the suspension. The migration of the roll centers and pitch centers could also be taken into account.

Balkwill, J., 2013, Advanced Chassis Engineering: MSc Motorsport Engineering: Oxford, Oxford Brookes University, 363 p.

Dominy, J. A., and R. G. Dominy, 1984, Aerodynamic influences on the performance of the Grand Prix racing car: The Institution of Mechanical Engineers, v. 198.

Tremblay, A., 2013, Steady-State Laptime Simulation, Oxford.